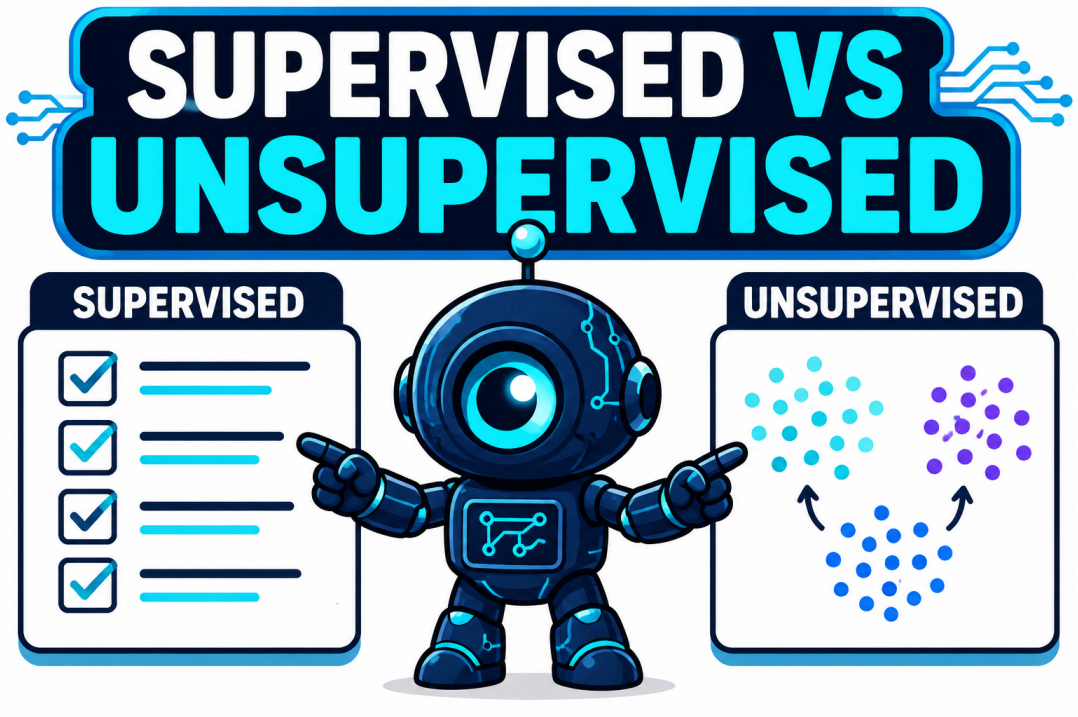
AI MECH LAB

AI Course — Beginner Track



Module 4, Topic 4.1

Supervised vs. Unsupervised Learning

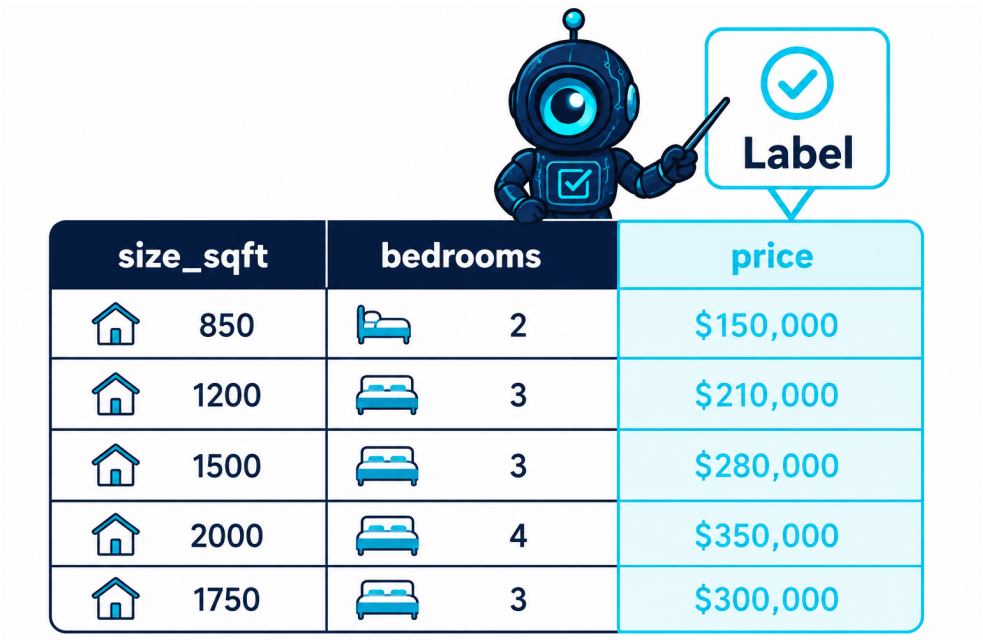
Lesson

You've covered the math and the tools. Now you start the part of this course that's actually about machine learning itself — and the first idea you need is the single biggest fork in the road: does your model learn from labeled answers, or does it have to find structure on its own? That fork is supervised versus unsupervised learning, and almost everything else in ML sits on one side of it or the other.

What makes learning "supervised"?

In supervised learning, every example in your training data comes with the correct answer attached. You're not just handing the model raw numbers — you're handing it raw numbers *and* the outcome you want it to learn to predict.

Take a table of houses:



Features (size, bedrooms) paired with a known label (price).

size_sqft and bedrooms are the features — the inputs the model looks at. price is the label — the known, correct answer for each row. The model's whole job during training is to find the pattern connecting features to label, closely enough that it can guess correctly on a house it's never seen.

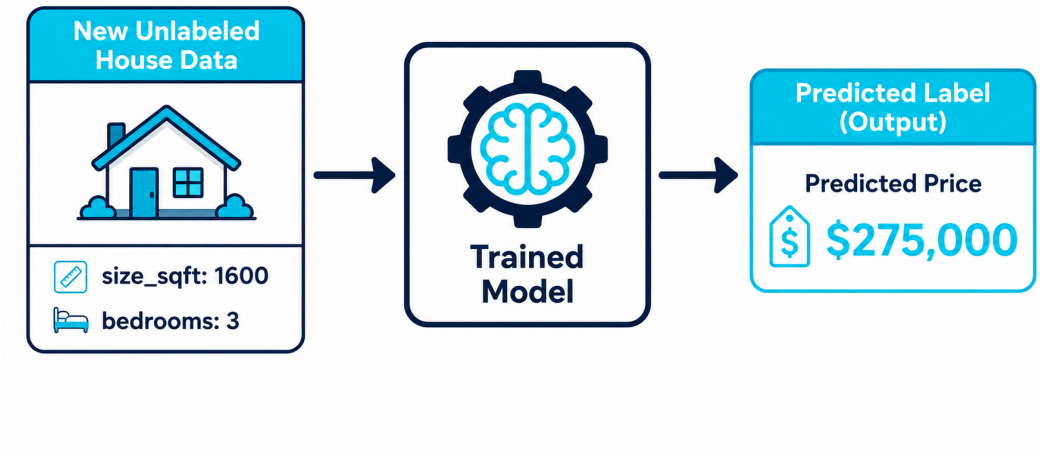
This is "supervised" in a literal sense: a human already worked out the right answer for every training example, and the model learns under that supervision.

How a trained model actually gets used

Once training is done, the process for a brand-new house looks like this: you feed it fresh, unlabeled feature values — size and bedroom count for a house the model has never seen — and the trained model outputs a predicted label. No price was given going in; the model produces one coming out.

```
new_house = {"size_sqft": 1600, "bedrooms": 3}
predicted_price = model.predict(new_house)
```

```
# predicted_price ~ $275,000
```



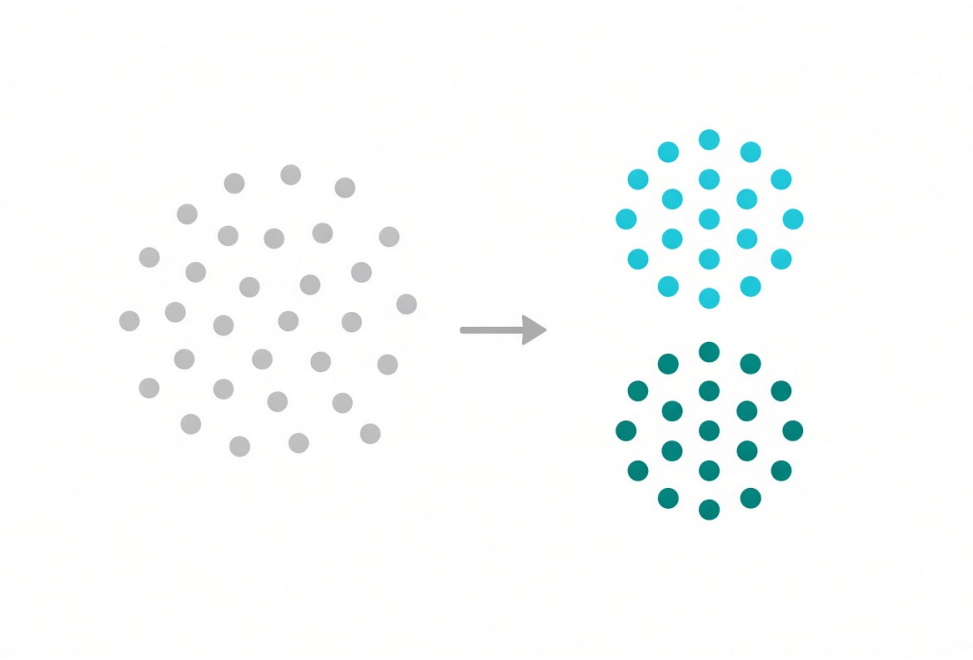
A trained model turns unlabeled input into a predicted label.

That's the entire point of supervised learning: train on examples where you already know the answer, so the model can predict the answer for cases where you don't.

What happens when there are no labels at all

Unsupervised learning starts from a completely different position: you have data, but nobody has attached correct answers to any of it. There's no "price" column, no known category, nothing to supervise the model with.

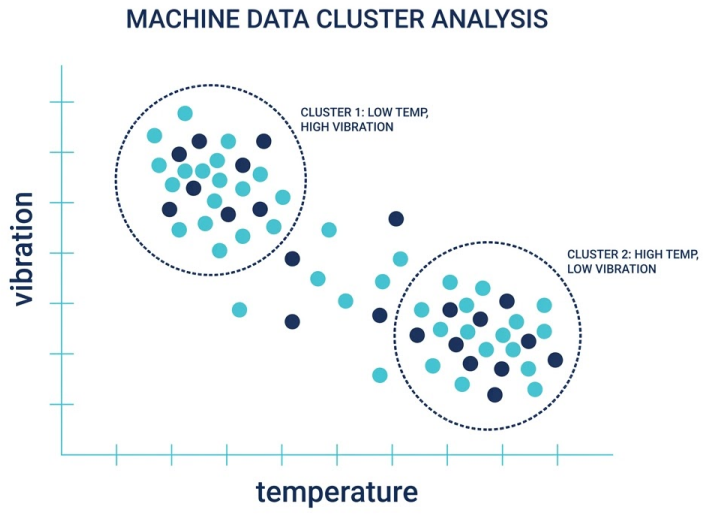
Instead of predicting a label, the model's job is to find structure that was already hiding in the data — grouping similar things together without ever being told what the groups should be. A pile of unlabeled points, run through an unsupervised algorithm, comes out sorted into clusters based purely on how similar the points are to each other.



Unlabeled points, grouped into clusters with no labels given.

A real-world example: machine sensor data

Say you're logging vibration and temperature readings off a set of machines on a factory floor, with no labels at all — nobody has marked which readings mean "healthy" or "failing." Plot vibration against temperature and an unsupervised algorithm can still find two distinct clusters: one group sitting at low temperature and high vibration, another at high temperature and low vibration.

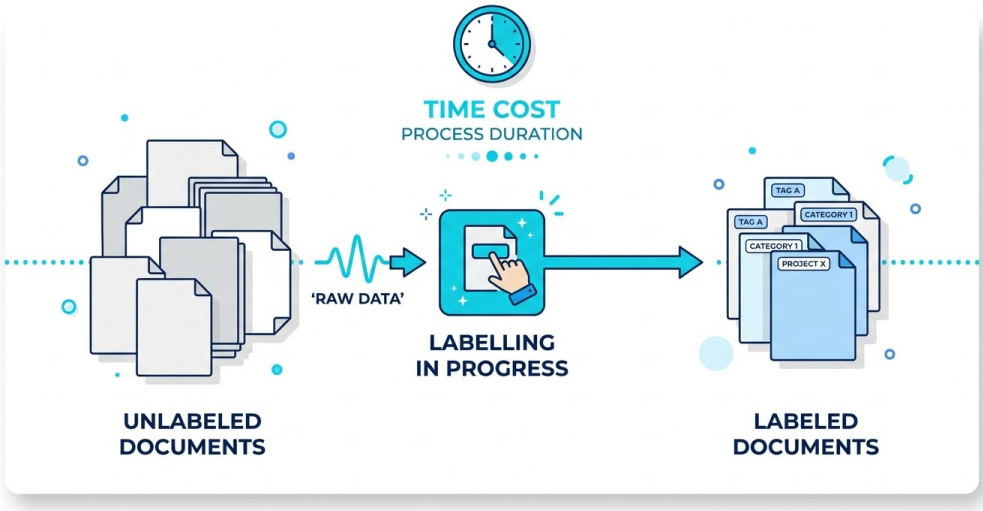


Two clusters emerge from unlabeled sensor readings.

Nobody told the algorithm what those clusters mean. It found the grouping on its own, purely from how the data points sit relative to each other. It's now up to you — the engineer — to look at each cluster and interpret what it represents, maybe by pulling a few machines from each group and inspecting them.

Why unsupervised learning exists — the real cost of labels

If labels are so useful, why not just label everything? Because labeling is expensive. Someone — a person — has to go through raw, unlabeled data and manually attach the correct tag to each item: reading every document and assigning it a category, inspecting every image and marking what's in it, checking every transaction and flagging it as normal or fraudulent. That takes real time and real cost, and at large scale it's often not realistic at all.



Turning unlabeled data into labeled data takes real time and cost.

Unsupervised learning matters because in a huge amount of real-world data, no one has ever gone through and labeled it. If you want any insight at all — structure, groupings, anomalies — before you've paid the cost of labeling, unsupervised methods are how you get it.

Supervised vs. unsupervised, side by side



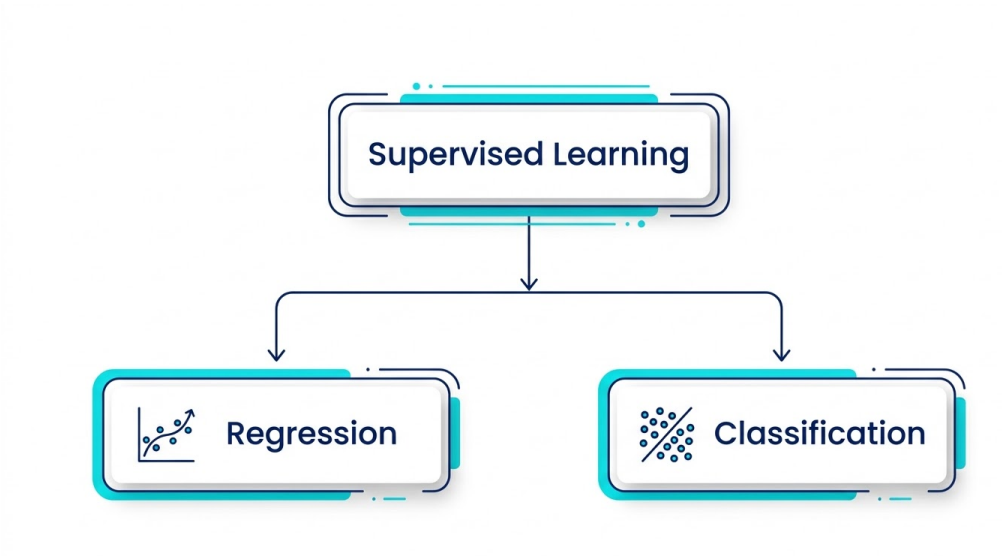
The core distinction: predicting a known outcome vs. finding unknown patterns.

	Supervised	Unsupervised

Data	Labeled	Unlabeled
Goal	Predict a known outcome	Find unknown patterns
Output	A specific predicted value or category	Groupings or structure
Example	Predicting house price	Grouping similar machines

Where supervised learning splits further

Supervised learning itself branches into two families, and you'll spend real time on both of them soon:



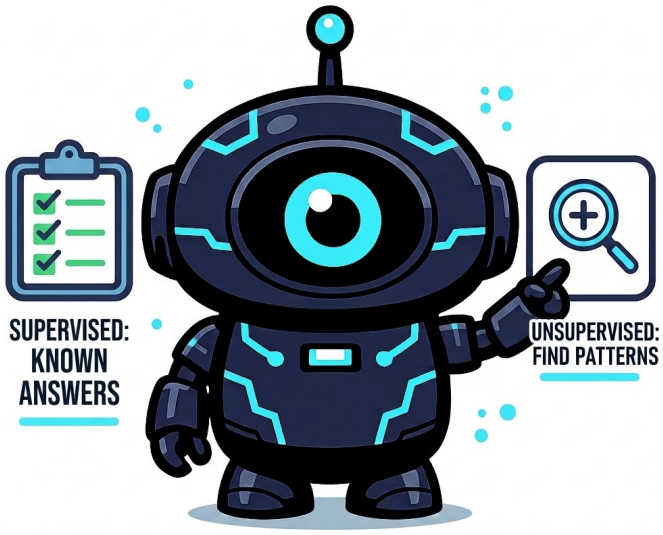
Supervised learning splits into Regression and Classification.

Regression — the label is a continuous number (price, temperature, distance).
Classification — the label is a category (cat/dog/bird, healthy/faulty, fraud/not fraud).

You'll cover each of these as its own topic next — this lesson is only building the frame they both sit inside.

Where this leaves you

By the end of this lesson, you should be able to look at any dataset and ask one question first: does this data have known correct answers attached, or not? If it does, you're in supervised territory, working toward predicting a label. If it doesn't, you're in unsupervised territory, working toward discovering structure. The next lesson zooms into the first branch of supervised learning — Regression — where you'll build your first real model that predicts a number.



The two core paths: known answers, or find patterns.